Unit: Unit 8: Quadratics **Name:** ________________
Topic: Definition of Quadratics and Standard Form **Date:** __________

Foundation (Jalapeno)

Q1. What is the standard form of a quadratic function?
(A) $ax^2 + bx + c$
(B) $ax + bx^2 + c$
(C) $ax^2 - bx + c$
(D) $ax + b$
(E) $ax^2 + c$

Q2. Which of the following is a quadratic function in standard form?
(A) $x^2 + 3x - 2$
(B) $x^2 - 3x$
(C) $x^2 + 3$
(D) $x + 3$
(E) $x^3 + 2x^2$

Q3. What is the value of a in the quadratic function $2x^2+3x-1$?
(A) 1
(B) 2
(C) 3
(D) -1
(E) -2

Q4. Which part of the quadratic function $ax^2 + bx + c$ is the constant term?
(A) ax^2
(B) bx
(C) c
(D) a
(E) b

Q5. What is the coefficient of the x^2 term in $-2x^2 + 3x - 1$?
(A) 1
(B) -1
(C) 2
(D) -2
(E) 3

Q6. Which of the following is NOT a quadratic function?
(A) $x^2 + 2x + 1$
(B) $x^2 - 3x$
(C) $x + 2$
(D) $x^2 + 1$
(E) $x^3 + 2x^2$

Q7. Identify the quadratic function: $f(x) =$
(A) $x^2 + 2x - 3$
(B) $x + 2$
(C) $x^2 - 2x + 1$
(D) $x^3 + 2x^2$
(E) $x^2 + 1$

Q8. What is b in $x^2 + 4x + 4$?
(A) 1
(B) 2
(C) 4
(D) -4
(E) -1

Open Ended Questions (FRQ)

Q9. Write a quadratic equation in standard form with $a = 2$, $b = -5$, and $c = 3$. Show your work and final equation.

Q10. Describe the key features of the quadratic function $f(x) = -x^2 + 2x - 1$. Be sure to include the x- and y-intercepts.

Chef's Tips

- Tip 1: Make sure students understand the standard form of a quadratic function.
- Tip 2: Encourage students to identify key features of quadratic functions.
- Tip 3: Remind students to show all work for free-response questions.